

SUBASH VELUMANI
105 Lake Village of Boulevard, Apartment#105, Dearborn, MI 48120
Email subashv9594@gmail.com | Phone +1 (313) 985-2176
LinkedIn <https://www.linkedin.com/in/subash-velumani-aaa546104>

OBJECTIVE

Seeking a role in **Quality and Manufacturing** with emphasis on continuous improvement, quality problem resolution and process development with an insight into achieving the company’s mission

SUMMARY

- Automotive engineer with **over 4 years of experience working in Quality and Supplier Quality**.
- Working closely with cross functional teams to provide problem resolution for quality issues.
- Applied working knowledge of product design and part assembly into the manufacturing processes.
- Implemented and maintained cost effective manufacturing processes with **ISO, IATF quality standards**.
- Coordinated new manufacturing launch for vehicle program at assembly plant.

TECHNICAL SKILLS AND CERTIFICATIONS

Certification	Lean Six Sigma Green Belt, GD &T, CNC Machining, PRO-E CREO V2.0
Skills	Problem Solving, Root Cause Analysis (8D, 5-Why, Fish Bone Diagram), FMEA, PPAP, APQP, Control Plans, MSA, Gage R&R, DMAIC, SPC, Process Capability, PDCA, Poke-Yoke Process Mapping, 5S, Kaizen.
Tools	Minitab, Solid works, AutoCAD, ERP SAP Crystal, MS Office Suite (Word, Excel, and PowerPoint)

WORK EXPERIENCE

Currently enrolled in school for MBA

Epitex Inc., Ford Motor Company Flat Rock, MI June 2023 – Oct 2024
Resident STA Engineer

- Involve in daily plant quality meeting, weekly Top 20 VRT meeting and monthly **Vehicle Quality Review (VQR)** meeting to provide support and update for supplier related quality issues on containment action if needed.
- Arrange meetings on **critical supplier quality issues** with cross-functional teams to drive the issues to closure.
- Review and Update **BSAQ (Balanced Single Agenda for Quality)** system with supplier 8D with **proper ICA and PCA** cut-off date to drive supplier quality issues to closure.
- Analyse **warranty claim report** and customer feedback to identify proper root-cause for binning.
- Working with **Incoming Quality (IQ)** for non-responsive suppliers for issue dispute and resolution.
- Working with IQ to initiate **3rd party in-house** or at supplier facility for **interim and permanent containment** for supplier quality issues.
- Working with **Site STA team** on critical supplier quality issues that needed **supplier plant visit** to check and verify their process and provide confirmation of the standards and quality procedures.
- Initiate **supplier PPAP reviews** for upper management for critical supplier quality issues and proper containment and preventive action documentation maintenance.
- Participate in management reviews as required - **NAQM (North America Quality Review), SARs (Special Attention Reviews)** and provide support for supplier related quality issues if needed.
- Collaborate with **Interior, Exterior and Electrical CFT's** to support supplier quality issues when needed.

Resource Tek., Mercedes Benz U.S. International Vance, AL Oct 2020 – June 2023
Part Quality Engineer

- Working with suppliers in lead **problem solving** activities for part quality issues using **5-Why (Why Ship Why Made) Root Cause Analysis** and collaboratively implementing Corrective and Preventive Action (**CAPA**) using **IRA 8D** containment report.
- Working closely with cross-functional teams (operations, design, analysis, supplier quality, process engineering, purchasing, logistics) in brainstorming meetings and resolving quality issues through **APQP** and **PPAP** manuals (**PFMEA, Control Plan, and Process Flow Diagrams**).

- Responsible for **exterior commodity** includes Glazing Systems (windshield, rear glass, side glass, quarter glass), ornamental exterior trims, sealings, wiper washer system, door handles, and exterior lighting.
- Involving in **new electric vehicle (EVA) model launch** (X296, X294, Z296, Z94) and **pilot** vehicle manufacturing.
- Leading weekly supplier **KTMA** reviews with design, analysis, supplier teams to analyse part dimension, measurement data and **Gage R&R** variability through **Pi-web (online)**, **DIALOG (2D drawings)** and **Engineering client software (3D drawings)**
- Maintaining commodity quality issues through **SQMS** and involving in **cost and PPM reduction activities**.
- Tracking and monitoring daily online **FTC (Throughput)** and **CPU** quality issues through **MRS** and **Q-live** and generate reports if necessary.
- Providing timely updates and feedback of quality issues to **GFP** and **VOCA (Voice of Customer Audit)** management.

Sakthi Automotive Group Detroit, MI (Co/Op)
Quality Engineer

May 2017 – Sep 2020

- Lead **problem solving** activities for customer quality complaints using **Fishbone** diagram and **5-Why Root Cause Analysis** and collaboratively implemented Corrective and Preventive Action (**CAPA**) using **8D** containment report.
- Worked closely with cross functional teams and quality assurance team in updating **APQP** and **PPAP** manuals (**PFMEA, Control Plan, and Process Flow Diagrams**)
- Interfaced with internal quality auditor to audit **EHS ISO 14001:2004, TS 16949:2016** and Bureau of International Quality Standards (BIQS) audit for quality system regulations.
- Responsible to **Gage R&R** studies and **CMM inspections** for different projects.
- Responded to quality problems in the shop floor, created **rework instructions** and eliminated issues by collaborating with engineering and manufacturing teams.
- Performed **Layered Process Audits (LPA)** daily to ensure no uncertified stock.
- Developed and maintained all shop floor activities using **Statistical Process Control** Check Sheets
- Performed **Process Mapping** and **Capability Studies** to improve **First Time Throughput (FTT)** capability by eliminating bottlenecks and ensure adherence to customer specifications.
- Created and maintained **work instructions** and **Quality Alerts**. Trained operators to standardize work.
- Performed **cost reduction activities** by calculating **PPM** of non-conformance parts for customer review.
- Updated **PFMEA and Control Plans** through customer quality and shop floor quality concerns as per the findings.

EDUCATION

Master of Business Administration, Trine University IN

Aug 2025 – Present

Coursework: Strategic Marketing Management, Executive Communication, Managerial Accounting for MBA, Project Management

Master's in industrial engineering, Wayne State University MI

Sep 2015 - May 2017

Coursework: Quality Management system, Lean Operations and Manufacturing, Quality Engineering, Digital Automation, Deterministic Optimization, Supply Chain Management, Production Systems, Applied Engineering

Bachelor's in mechanical engineering, Anna University India

Aug 2011 - May 2015

Coursework: Automobile Engineering, Manufacturing Technology, Production Planning and cost estimation, Industrial Engineering and Operations Research, Meteorology Quality Management